

culture are important.

Dr Madhura Purnaprajna, Professor, PES University says, “We as a community are in need of leaders, people, researchers to drive this innovation. There is no dearth of talent, and the need is to facilitate innovation early while in the learning process.” According to her, innovation is a very tricky subject that has to be driven through careful thought, culture, and mindset. She adds, “I strongly believe that innovation is the key to drive self-reliance in the Indian scenario.”

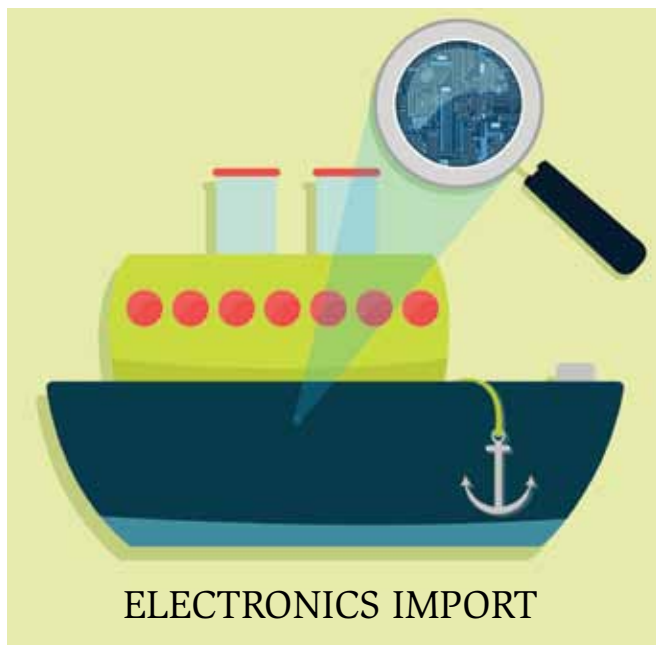
From employment perspective, electronics industry is very significant, especially with the country’s demand being projected at \$400 billion by 2025. The industry and academia need to come together as joint co-creators of innovation for self-reliance.

Experts believe this is right time for innovation in the electronics sector as compared to a decade ago. “We have platforms at the school level creating hobby electronics and getting people to actually do electronic development. In addition, these products are available so easily in the market that it’s quite easy for somebody to get in,” says Ravi Subramanian, Partner, ekLakshya Innovation Labs.

Collaborate and co-develop

Academia could play a larger role than merely being a supplier of manpower to the industries. Innovative results can be achieved through academic efforts. “Public and private collaborations are some directions to channelise these innovations,” says Purnaprajna.

From an academic perspective, the professor says, she has seen immense interest of the student community to participate in project-



driven initiatives. She thinks it would be great to move away from rote learning towards a problem-solving environment. “Industry people could come in earlier, maybe have an overlap of one or two years and motivate candidates. We tend to give them a theoretical perspective, while industry experts can give them a practical problem. This is how you can build socially-relevant solutions.”

Purnaprajna suggests this could be a better way than industry experts arriving at the end of fourth year for recruitments. Industry partners can participate in co-developing curriculum and projects along with students and faculty members in the long term. As universities become more open to making changes in curriculum, experts believe industry participation can take innovation forward. With industry coming into the picture early a real big difference can be made possible.

In contrast, G.S. Madhusudan, Founder & CEO, Incore Semiconductors, says initiatives and programmes collapse in absence of institutional drive. “A bachelor’s course in electronics means the candidate should have at least

got some experience in designing a practical SMPS, audio amplifier, CPU kit, and probably one more design either in digital or analogue. These, when built practically, tie the theory and practice. So, even if the prototype does not succeed, it allows us to teach how to analyse what went wrong or right, which leads candidates to debug or figure out their wrongs and create success stories in future. This is engineering, where you go through the whole process of reasoning and analysis of building systems. So, there needs to be a differ-

ence in the approach and method of teaching.

At present, when a student transitions from university to industry, the initial months are unproductive. Madhusudan thinks there should be an apprenticeship programme for a year after the four-year academic study, in place of short-term internships. “Despite the fact that all kits are available as open source, there is a lack of guidance,” he says.

Going beyond research

For innovation, research is an absolute necessity. Our institutions already have an ecosystem that focuses on research. So, industry-academia collaboration could inject new scientific ideas into industry.

There are quite a few institutions that are keen on getting into product development. They could take it to the next level by supporting startups with some money based on their ideas. “We are very uniquely placed in terms of getting the next-level BB ecosystem, getting eased up as well as the government initiatives pushing us in terms of innovation in the electronics sector. There is a lot of opportunity for the students who look at developing prototypes, or as